SALES ANALYTICS / BUSINESS INTELLIGENCE

POWER BI / PORTFOLIO CASE STUDY

Sales Analytics Dashboard

Business Performance & Profitability Review | Reporting period: 2024–2025

Executive brief

This report evaluates commercial performance through a four-page Power BI dashboard covering sales, product mix, customer segments, regional profitability and month-over-month trends. It translates descriptive metrics into business questions and proposed actions. Findings are based on the dashboard screenshots and the monthly values checked during development; the transaction-level source has not been independently audited.

Performance at a glance

SALES	PROFIT	MARGIN	CUSTOMERS
1.24M	353.81K	28.62%	298

LATEST MONTH	PREVIOUS MONTH	MONTHLY CHANGE
Dec 2025 · 56,325.07	Nov 2025 · 47,557.87	+18.43% (+8,767.20)

Executive takeaways

- Technology accounts for the largest share of revenue; Laptop is the leading individual product by sales.
- Consumer contributes the largest absolute profit among the three customer segments.
- Regional profit margins appear close (roughly 28–29%), despite differences in sales volume.
- December 2025 sales recovered versus November, but the series also shows a pronounced low in March 2025 requiring investigation.

Scope & decision questions

The analysis addresses four questions: How is overall performance evolving? Which products and categories drive results? Which customers and regions contribute profit? Where should management investigate performance changes?

01) Data & analytical approach

Data model and preparation

The Power BI model uses a Sales fact table and a Calendar date table, linked through Calendar[Date] → Sales[Order_Date]. Power Query was used to review data types, duplicate handling and arithmetic consistency. A temporary check tested whether Profit = Sales − Cost; the user confirmed that the checked results were zero after correcting numeric types.

Measures and definitions

Metric	Definition / implementation note
Total Sales	Aggregate of sales amounts.
Total Profit	Aggregate profit; displayed as approximately 353.81K.
Profit Margin	Profit divided by sales; displayed as 28.62%.
Previous Month Sales	Time-shifted sales using DATEADD(Calendar[Date], −1, MONTH).
Monthly Sales Growth	Change relative to the previous month; the grand-total value is not the latest-month rate.
Latest Month Sales / Growth	December 2025 values shown in dedicated KPI cards.

Data coverage and quality

- Two calendar years are shown: January 2024 through December 2025.
- Nine distinct product names were verified. The visual is correctly titled “Sales by Product,” not “Top 10 Products.”
- The currency of the monetary values has not been established. Monetary amounts are intentionally presented without a currency symbol.
- KPI cards rounded to K/M should not be treated as exact values. Detailed monthly values are used when available.

02) KPI register

KPI	Dashboard value	Business interpretation
Total Sales	1,236,320.39	Total revenue proxy over the displayed period
Total Profit	~353.81K	Absolute profit
Profit Margin	28.62%	Overall profitability ratio
Total Orders	~2K	Rounded transaction / order count
Total Customers	298	Customer base represented
Average Order Value	824.21	Mean sales per order, per model measure
Average Discount	5.41%	Discount indicator
Latest Month Growth	+18.43%	December 2025 vs November 2025

03) Findings by dashboard page

Executive Overview

The monthly sales line reveals uneven performance rather than steady growth. A particularly weak month is visible in March 2025. Technology leads category sales (approximately 0.77M), ahead of Furniture (~0.44M) and Office Supplies (~0.03M). The executive page supports rapid monitoring of total sales, profit, orders and margin, with region and segment filters.

Product & Sales Analysis

Laptop contributes approximately 0.57M in sales, followed by Desk (~0.24M) and Monitor (~0.16M). These figures indicate a concentrated product mix. Technology leads category profit (~190K on the dashboard), followed by Furniture (~149K), while Office Supplies contributes substantially less (~15K). These are displayed rounded values and should be confirmed against underlying records before external business decisions.

Customer & Profitability Analysis

The dashboard represents 298 customers and an average order value of 824.21. Consumer contributes the highest absolute profit (approximately 127K), ahead of Corporate (~117K) and Home Office (~111K). The regional margin chart shows relatively narrow variation, roughly 28–29%, suggesting that sales volume and profit rate should be assessed together.

Advanced Sales Insights

December 2025 sales total 56,325.07, compared with 47,557.87 in November. The increase is 8,767.20, or +18.43%. The two-line chart compares current-month and previous-month sales across the 2024–2025 period. The 4.77% grand-total growth shown in the detailed table is a calculation over the full evaluation context; it must not be described as December growth.

Interpretation guardrails

- Category and segment leaders are identified by absolute value, not necessarily by efficiency or return on investment.
- The March 2025 decline is an observation; its cause is not established by the dashboard alone.
- Comparisons by region or segment depend on slicer interactions and should be validated in the PBIX file.

04) Recommendations & next actions

Priority	Action	How to validate impact
High	Investigate the March 2025 sales decline by product, region, segment, and order count.	Compare monthly breakdowns and identify the largest negative contributors.
High	Monitor the leading Technology/Laptop concentration.	Track share of sales, category margin and changes over time.
Medium	Review lower-volume products and office Supplies.	Compare units, discounts, revenue and profit before recommending changes
Medium	Pair regional revenue with regional margin in reviews.	Assess whether high-sales regions also produce proportionate profit.
Medium	Use a monthly performance review with current, previous and growth metrics.	Track month-over-month changes and investigate material deviations.

05) Limitations & validation checklist

- This report was assembled from dashboard screenshots and previously checked monthly figures, not a full transaction-level audit.
- Confirm the currency, source dataset, refresh period and whether public redistribution is permitted before publishing.
- Test Region and Segment slicers on every page, including “latest month” measures that may intentionally ignore filters.
- Verify the chronological order of the Advanced Sales Insights table and ensure exported screenshots show the intended sort.
- Confirm the exact order count and category/segment totals from Power BI if publishing them as precise figures.

06) Portfolio deliverables

Asset	Purpose
Sales_Analytics_Dashboard.pbix	Editable Power BI report and model
README.md	Project objectives, workflow, metrics, findings and usage
Dashboard screenshots	Visual overview of each of the four pages
Business analysis report	Interpretation, recommendations and documented limitations

Conclusion

This project demonstrates an end-to-end business intelligence workflow: data preparation, date modeling, DAX time intelligence, interactive visualization and decision-oriented interpretation. The strongest business value lies in using the dashboard to investigate performance drivers rather than merely reporting headline totals.